

Ans. to the Q.NO - 101)

(i) We know that, the time per taken to travel from one point of zero velocity to the next such point is half of the total time period of a complete oscillation. ( $\frac{T}{2} = t$ )

$$\begin{aligned}\therefore T &= 2t \\ &= 2 \times 0.25 \text{ (given, time } t = 0.25\text{s)} \\ &= 0.5\text{s}\end{aligned}$$

(ii) Frequency,  $f = \frac{1}{T} = \frac{1}{0.5} = 2\text{ Hz}$

(iii)  $x_{\max} = A = \frac{36}{2} = 18\text{ cm}$

Ans. to the Q.NO (02)

given, mass,  $m = 680 \text{ gm} = 0.68 \text{ kg}$

spring constant,  $K = 65 \text{ Nm}^{-1}$

Maximum displacement,  $x = A = 11 \text{ cm} = 0.11 \text{ m}$

(i) angular frequency,  $\omega = \sqrt{\frac{K}{m}}$   
 $= \sqrt{\frac{65}{0.68}}$   
 $= 9.8 \text{ rad s}^{-1}$

(i) Time period,  $T = \frac{2\pi}{\omega}$   
 $= \frac{2\pi}{9.8}$   
 $= 0.64 \text{ s}$

$$T = 2\pi \times \sqrt{\frac{m}{K}}$$
$$= 2\pi \times \sqrt{\frac{0.68}{65}}$$
$$= 0.64 \text{ s}$$

(iii)  $\phi = \delta = \cos^{-1}\left(\frac{x_0}{A}\right) = \cos^{-1}\left(\frac{A}{A}\right) = \cos^{-1}(1) = 0^\circ$

(iv) Velocity at any time:  $v(t) = -A\omega \sin(\omega t + \phi)$   
 $v(4s) = -(0.11)(9.8) \sin(9.8 \times 4 + 0)$   
 $= -1.075 \text{ m/s}$

$$V_{\max} = \omega A$$

$$= 9.8 \times 0.11$$

$$= 1.078 \text{ m/s}$$

(v) acceleration at any time:  $a(t) = -A\omega^2 \cos(\omega t + \phi)$   
 $a(4s) = -(0.11)(9.8)^2 \cos(9.8 \times 4 + 0)$   
 $= -0.73 \text{ m/s}^2$

$$a_{\max} = -\omega^2 A$$

$$= -(9.8)^2 (0.11)$$

$$= -10.56 \text{ m/s}^2$$

(vi) Displacement at any time:  $x(t) = A \cos(\omega t + \phi)$   
 $x(0s) = 0.11 \cos(\omega \times 0 + 0)$   
 $= 0.11 \text{ m}$   
 $x(7s) = 0.11 \cos(9.8 \times 7 + 0)$   
 $= 0.095 \text{ m}$

(viii) Velocity at any displacement:  $v(x) = \omega \sqrt{A^2 - x^2}$   
 $v(0.11 \text{ m}) = 9.8 \sqrt{(0.11)^2 - (0.11)^2}$   
 $= 9.8 \times 0$   
 $= 0 \text{ m/s}$   
 $v(0.04 \text{ m}) = 9.8 \sqrt{(0.11)^2 - (0.04)^2}$   
 $= 1.004 \text{ m/s}$

Ans. to the Q.No (09)

given, mass,  $m = 0.12 \text{ kg}$

Amplitude,  $A = 8.5 \text{ cm} = 0.085 \text{ m}$

Time period,  $T = 0.20 \text{ s}$

(i)

$$F_{\max} = ma$$

$$= m \omega^2 x$$

$$= m \omega^2 A$$

$$= 0.12 \times (31.416)^2 \times 0.085$$

$$= 10.07 \text{ N}$$

$$\omega = \frac{2\pi}{T}$$

$$= \frac{2\pi}{0.20}$$

$$= 31.416 \text{ rad/s}$$

(ii)

We know,  $k = \frac{F}{x}$

$$= \frac{10.07}{0.085}$$
$$= 118.5 \text{ Nm}^{-1}$$

$$k = \omega^2 m$$

$$= (31.416)^2 (0.12)$$

$$= 118.5 \text{ Nm}^{-1}$$

Ans. to the Q.No(04)

$$f = 120 \text{ Hz}$$

$$\text{distance, } x = 2 \text{ mm} = 0.002 \text{ m}$$

$$\begin{aligned} \text{(i)} \quad A &= \frac{x}{2} \\ &= \frac{2}{2} \\ &= 1 \text{ mm} \\ &= 0.001 \text{ m} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad V_{\max} &= \omega A \\ &= (240\pi)(0.001) \\ &= 0.75 \text{ ms}^{-1} \end{aligned}$$

$$\begin{aligned} \omega &= 2\pi f \\ &= 2\pi \times 120 \\ &= 240\pi \text{ rad s}^{-1} \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad a_{\max} &= \omega^2 A \\ &= (240\pi)^2 (0.001) \\ &= 568.4 \text{ ms}^{-2} \end{aligned}$$

Ans. to the Q.No(05)

$$F = -Kx$$

$$ma = Kx$$

$$\text{given, } K = 400 \text{ N m}^{-1}$$

$$x = 0.100 \text{ m}$$

$$v = 13.6 \text{ m s}^{-1}$$

$$a = 123 \text{ m s}^{-2}$$

$$(i) \quad K = \frac{F}{x}$$

$$ma = Kx$$

$$m = \frac{Kx}{a} = \frac{400 \times 0.1000}{123} = 0.325 \text{ kg}$$

$$(i) \quad f = \frac{1}{T} = \frac{\omega}{2\pi} = \frac{1}{2\pi} \times \sqrt{\frac{K}{m}}$$
$$= \frac{1}{2\pi} \times \sqrt{\frac{400}{0.325}}$$
$$= 5.54 \text{ Hz}$$

$$\omega = 34.81$$

$$(ii) \quad v = \omega \sqrt{A^2 - x^2}$$

$$\frac{v}{\omega} = \sqrt{A^2 - x^2}$$

$$\left(\frac{v}{\omega}\right)^2 = A^2 - x^2$$

$$\sqrt{\left(\frac{v}{\omega}\right)^2 + x^2} = A$$

$$\sqrt{\left(\frac{13.6}{34.81}\right)^2 + (0.100)^2} = A$$

$$0.4003 \text{ m} = A$$

$$\omega = \sqrt{\frac{K}{m}}$$

$$= \sqrt{\frac{400}{0.325}}$$

$$= 35.08 \text{ rad s}^{-1}$$

Ans. to the Q. No (06)

$$m = 25 \text{ gm} = 0.025 \text{ kg}$$

$$K = 400 \text{ dyn/cm} = 40 \text{ Nm}^{-1}$$

$$x = A = 10 \text{ cm} = 0.1 \text{ m}$$

$$\begin{aligned} \text{(i)} \quad T &= \frac{2\pi}{\omega} \\ &= 2\pi \times \sqrt{\frac{m}{K}} \\ &= 2\pi \times \sqrt{\frac{0.025}{40}} \\ &= 0.157 \text{ s} \end{aligned}$$

$$\text{(ii)} \quad f = \frac{1}{T} = \frac{1}{0.157} = 6.37 \text{ Hz}$$

$$\begin{aligned} \text{(iii)} \quad \omega &= \sqrt{\frac{K}{m}} \\ &= \sqrt{\frac{40}{0.025}} \\ &= \cancel{0.16} \times 40 \text{ rad/s} \\ &\quad \text{Ans} \end{aligned}$$

### Ans. to the Q.No (7)

We know,  $F = Kx$

$$K = \omega^2 m$$

$$\omega = 2\pi f$$

$$= 2\pi \times 10^4$$

$$= 6.28 \times 10^4$$

$$\left| \begin{array}{l} m = 1.68 \times 10^{-27} \text{ kg} \\ f = 10^4 \text{ Hz} \\ A = 10^{-10} \text{ m} = x \end{array} \right.$$

$$\therefore F = Kx$$

$$= \omega^2 m x$$

$$= (6.28 \times 10^4)^2 (1.68 \times 10^{-27}) (10^{-10})$$

$$= 6.63 \times 10^{-8} \text{ N}$$

$$\left| \begin{array}{l} F = ma \\ = m \omega^2 x \\ = m \omega^2 A \quad [x=A] \end{array} \right.$$

### Ans. to the Q.No (08)

$$v = 1 \text{ m s}^{-1}$$

$$a = 1.57 \text{ m s}^{-2}$$

$$v_{\max} = \omega A$$

$$\Rightarrow 1 = \omega A$$

$$\Rightarrow A = \frac{1}{\omega}$$

$$\text{Now, } a_{\max} = \omega^2 A$$
$$= \omega^2 \left( \frac{1}{\omega} \right)$$

$$\therefore a = \omega = 1.57 \text{ rad s}^{-1}$$

$$\therefore T = \frac{2\pi}{\omega} = \frac{2\pi}{1.57} = 4.002 \text{ s}$$



Ans. to the Q. NO (09)

$$a_{\max} = \omega^2 A$$

$$\Rightarrow \omega = \sqrt{\frac{a_{\max}}{A}}$$

$$= \sqrt{\frac{48}{5}}$$

$$= 3.09 \text{ rad s}^{-1}$$

$$\left| \begin{array}{l} A = 5 \text{ m} \\ x = 0 \\ a = 48 \text{ m s}^{-2} \end{array} \right.$$

$$v_{\max} = A\omega$$

$$= 5(3.09)$$

$$= 15.45 \text{ m s}^{-1}$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{3.09} = 2.03 \text{ s}$$

$$v = \omega \sqrt{A^2 - x^2}$$

$$= 3.09 \sqrt{5^2 - 3^2}$$

$$= 12.36 \text{ m s}^{-1}$$

Ans. to the Q. NO (10)

$$(i) T = \frac{1}{f} = \frac{1}{0.25} = 4 \text{ s}$$

$$\left| \begin{array}{l} x_{\max} = A = 0.0037 \text{ m} \\ f = 0.25 \text{ Hz} \end{array} \right.$$

$$(ii) \omega = 2\pi f = 2\pi(0.25) = 1.57 \text{ rad s}^{-1} = \frac{\pi}{2} \text{ rad s}^{-1}$$

$$(iii) A = x = 0.0037 \text{ m}$$

$$(iv) x = A \cos \omega t = 0.0037 \cos \left\{ \left( \frac{\pi}{2} \times 3 \right) \right\} = 0 \text{ m}$$

$$(v) v = -A\omega \sin \omega t$$

$$= 0.0037 \times \frac{\pi}{2} \sin \left( \frac{\pi}{2} \times 3 \right)$$

$$= 0.0058 \text{ m s}^{-1}$$

Ans. to the Q. NO (11)

$$x = 10 \cos \left( 3\pi t + \frac{\pi}{3} \right) \quad \left| \begin{array}{l} A = 10 \text{ m} \\ \omega = 3\pi \text{ rad/s} \end{array} \right.$$

$$(i) \quad x = 10 \cos \left( 3\pi \times 2.5 + \frac{\pi}{3} \right) = 8.66 \text{ m}$$

$$(ii) \quad v = -10(3\pi) \sin \left( 3\pi \times 3 + \frac{\pi}{3} \right) = -81.42 \text{ m/s}$$

$$(iii) \quad a = -A\omega^2 \cos(\omega t + \phi) \\ = -10(3\pi)^2 \cos \left( 3\pi \times 2 + \frac{\pi}{3} \right) \\ = -444.13 \text{ m/s}^2$$

$$(iv) \quad v_{\max} = A\omega = 10(3\pi) = 94.24 \text{ m/s}$$

Ans. to the Q. NO (12)

$$x = 10 \sin \left( 10t - \frac{\pi}{6} \right) \\ \therefore x = 10 \cos \left( 10t - \frac{\pi}{3} \right)?$$

$$(i) \quad f = \frac{1}{T} = \frac{\omega}{2\pi} = \frac{10}{2\pi} = 1.59 \text{ Hz}$$

$$(ii) \quad T = \frac{1}{f} = 0.628 \text{ s}$$

$$(iii) \quad x_{\max} = A = 10 \text{ m}$$

$$(iv) \quad v_{\max} = A\omega = 10 \times 10 = 100 \text{ m/s}$$

$$(v) \quad a_{\max} = a^2 A = 10(10)^2 = 1000 \text{ m/s}^2$$

13 Given,

$$i) T = \frac{1}{f} = \frac{1}{0.5} = 2 \text{ sec}$$

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{2} = \pi$$

$$ii) x = A \cos(\omega t) \\ = 4 \cos(\pi t)$$

$$iii) v = -A\omega \sin(\omega t) \\ = -4\pi \sin(\pi(5)) \\ = 0 \text{ m/s}$$

$$\begin{cases} A = 4 \text{ m} \\ f = 0.5 \text{ Hz} \end{cases}$$

$$iv) a = -A\omega^2 \cos(\omega t) \\ = -4(\pi)^2 \cos(\pi(5)) \\ = -39.47 \text{ m/s}^2$$

$$\begin{cases} k = 106 \text{ N/m} \\ m = 2 \text{ kg} \end{cases}$$

$$\underline{14} i) \omega = \sqrt{\frac{k}{m}} \\ = 7\sqrt{2} \text{ rad/s}$$

$$f = \frac{\omega}{2\pi} = \frac{7\sqrt{2}}{2\pi} = 1.57 \text{ Hz}$$

$$T = \frac{1}{f} = 0.63 \text{ sec}$$

$$ii) x = 0.05 \cos(7\sqrt{2} t)$$

$$\begin{cases} A = 0.05 \\ \omega = 7\sqrt{2} \end{cases}$$

Ans to the Q.No(15)

$$(i) F = kx$$

$$ma = kx$$

$$mg = kx$$

$$\Rightarrow k = \frac{mg}{x} = \frac{0.5 \times 9.8}{0.07} = 70 \text{ Nm}^{-1}$$

$$m = 500 \text{ gm} = 0.5 \text{ kg}$$

$$A = 7 \text{ cm} = 0.07 \text{ m}$$

$$x = 3 \text{ cm} = 0.03 \text{ m}$$

$$V = 40 \text{ cm s}^{-1} = 0.4 \text{ m s}^{-1}$$

$$g = 9.8 \text{ m s}^{-2}$$

$$(ii) \omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{70}{0.5}} = 2\sqrt{35} \text{ rad s}^{-1}$$

$$(iii) T = \frac{2\pi}{\omega} = \frac{2\pi}{2\sqrt{35}} = 0.5315$$

$$(iv) E_p = \frac{1}{2} kx^2$$

$$= \frac{1}{2} (0.07)^2 (70)$$

$$= 0.1715 \text{ J}$$

$$(v) E_k = \frac{1}{2} mv^2$$

$$= \frac{1}{2} (0.5) (0.4)^2$$

$$= 0.04 \text{ J}$$

$$(vi) \text{ Amplitude of } E = E_p + E_k$$

$$\frac{1}{2} kA^2 = 0.2115$$

$$\Rightarrow A = \sqrt{\frac{2 \times 0.2115}{0.07}} = 2.45 \text{ m}$$

$$E = K + U$$

$$= 0.1715 + 0.04$$

$$= 0.2115 \text{ J}$$

## Ans. to the Q. No (16)

$$\begin{aligned} (a) \omega &= \sqrt{\frac{k}{m}} \\ &= \sqrt{\frac{400}{2.5}} \\ &= 4 \text{ rad s}^{-1} \end{aligned}$$

$$\begin{aligned} m &= 25 \text{ gm} \\ k &= 400 \text{ dyn/cm} \\ A &= 10 \text{ cm} \\ V &= 40 \text{ cm s}^{-1} \end{aligned}$$

$$(b) f = \frac{1}{T} = \frac{\omega}{2\pi} = \frac{4}{2\pi} = 0.63 \text{ Hz}$$

$$(d) E = \frac{1}{2} k A^2 = \frac{1}{2} (400)(10)^2 = 20 \times 10^3 \text{ erg}$$

$$(e) A = 10 \text{ cm}$$

$$\begin{aligned} (g) V_{\max} &= -A\omega \\ &= -4 \times 10 \\ &= -40 \text{ cm s}^{-1} \end{aligned}$$

$$\begin{aligned} \Delta_{\max} &= -A\omega^2 \\ &= 4^2 \times 10 \\ &= 160 \text{ cm s}^{-2} \end{aligned}$$

Ans. to the Q.No (17)

$$(i) K = m\omega^2 r$$

$$= 2.72 \times 10^5 (2\pi \times 10)$$

$$= 1.073 \times 10^9 \text{ Nm}^2$$

$$m = 2.72 \times 10^5 \text{ Kg}$$

$$f = 10 \text{ Hz}$$

$$E = K + U$$

$$= \frac{1}{2} mv^2 + \frac{1}{2} Kx^2$$

$$= 0 + \frac{1}{2} (1.073 \times 10^9) (0.20)^2$$

$$= 2.147 \times 10^7 \text{ J}$$

$$(ii) E = K + U$$

$$2.147 \times 10^7 = \frac{1}{2} mv^2 + \frac{1}{2} Kx^2$$

$$= \frac{1}{2} (2.75 \times 10^5) v^2 + 0$$

$$\therefore v = 12.6 \text{ m/s}$$